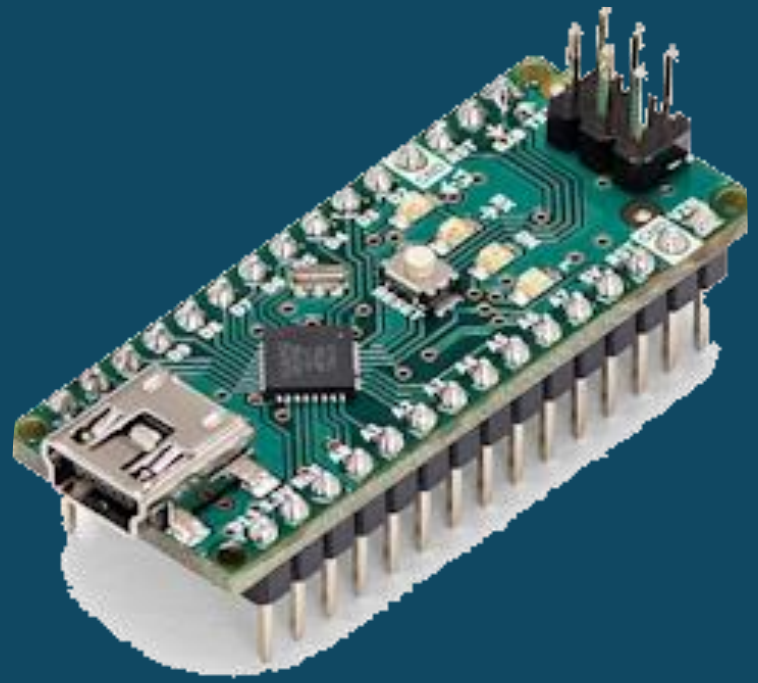


# UMRT Arduino Workshop

E2-351 6pm-7:30pm 2026-04-02

Quinn Gibbons & Quinn Thiessen

# Schedule

| | |
|---|---|
| Arduino 101 | What is the Arduino & some of it's uses<br>Basics of Arduino<br>Connecting components |
| Setup | Downloading the IDE<br>Board and Port selection<br>Installing Libraries |
| Coding 101 | Basics of the Arduino IDE<br>Setup of setup<br>Important Functions |
| Quick Project | We will do a simple project together<br>You will have time to try your own variations |
| Future Resources / Advice | Kits to buy & Resources for future learning<br>Last bits of Advice |

# Downloading the IDE

Go to this website: https://docs.arduino.cc/software/ide/

(on your laptop)

# What is the Arduino?

The hardware:

- A variety of microcontrollers with unique functions, allowing you to connect to and control real world components.

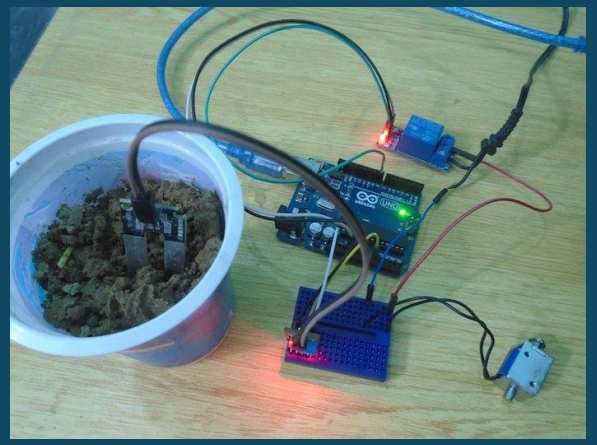
The software:

- The Arduino IDE uses a language similar to C++.

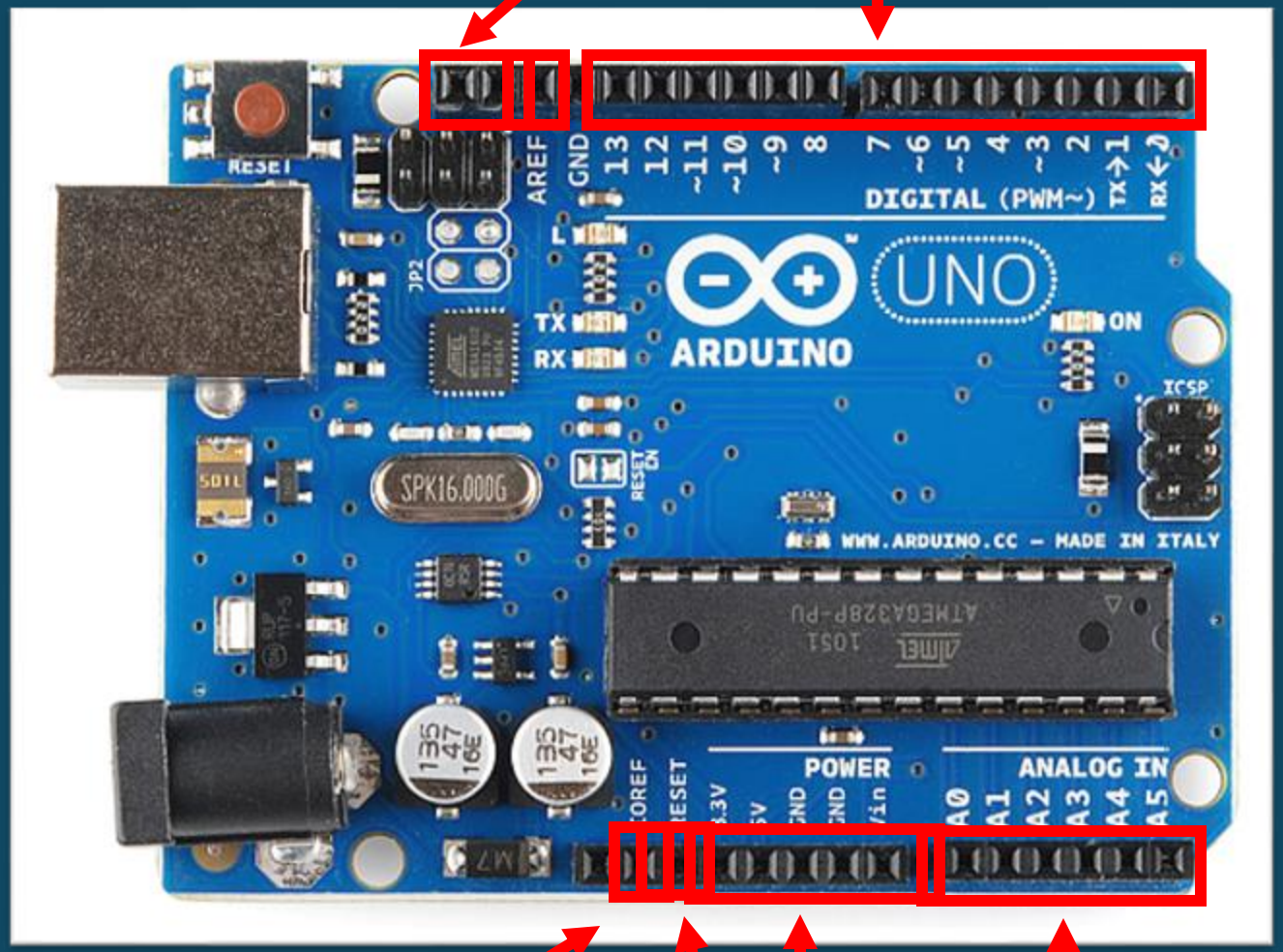
The community:

- Everything is open source, meaning offbrand boards can legally be reproduced by anyone. This also means there is a massive shared library for Arduino components as a resource.

# Somethings you can do…



# Basics of the board



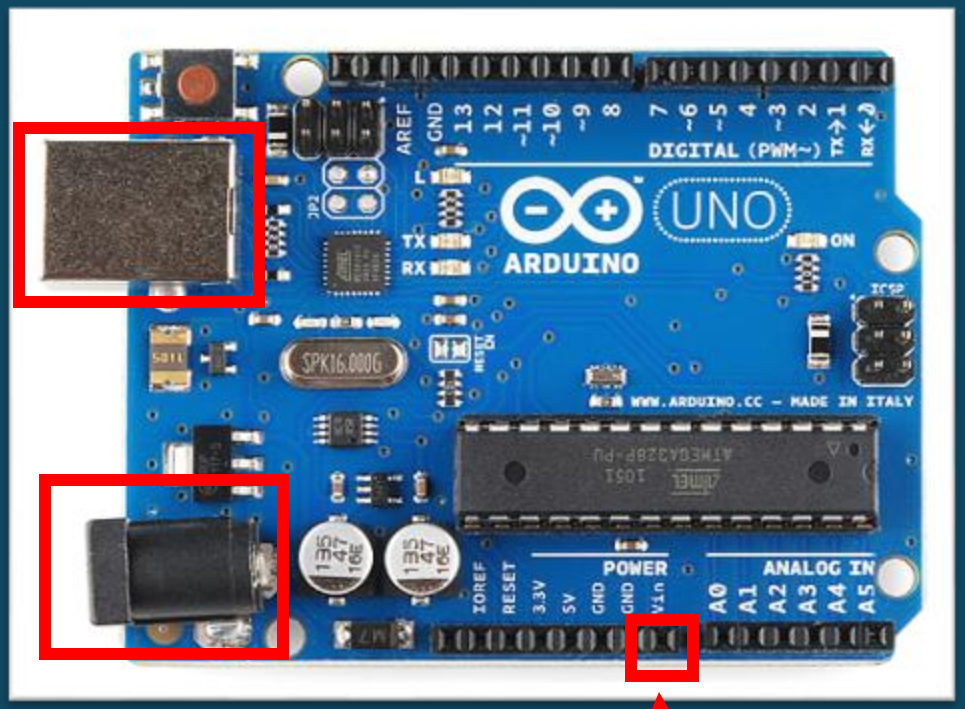

- Note: if they have a ~, they are capable of pulse width modulation (PWM)

# Powering the Arduino

- It is recommended to power the Arduino via USB port to your computer

USB port (5V)

DC Barrel Jack (7-12V)

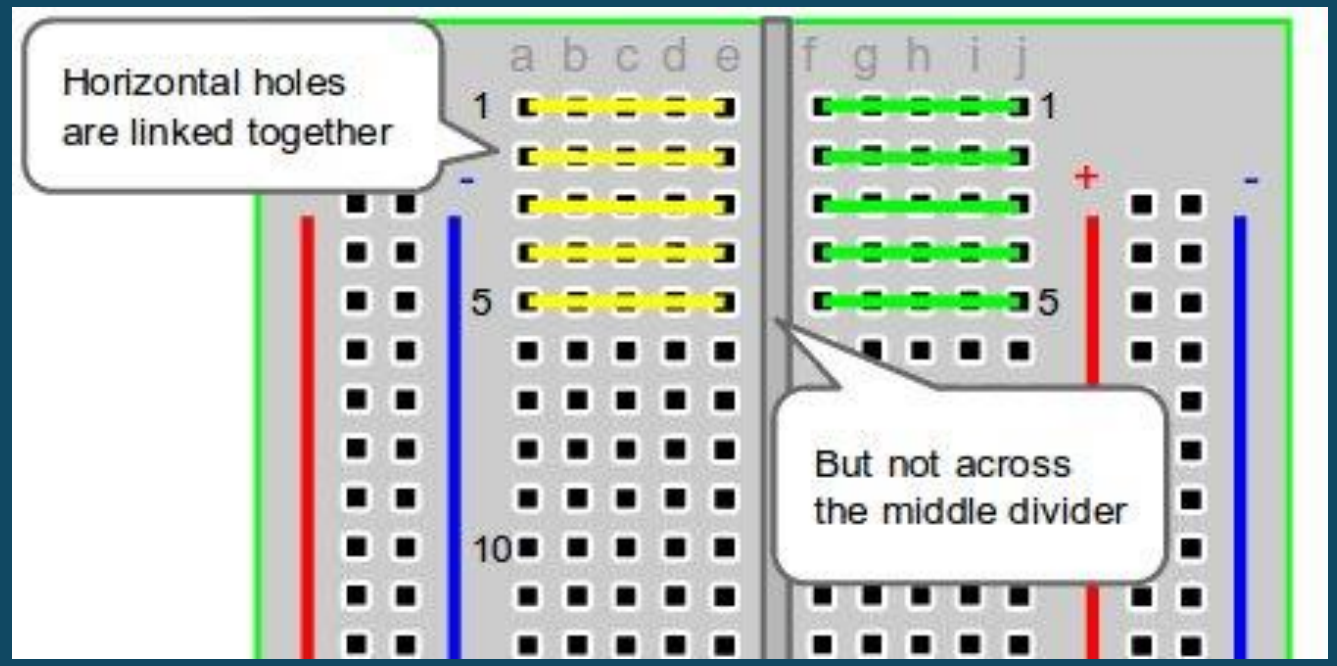



Vin port (7-12V)

# Breadboard

The breadboard will allow you to build circuits without soldering.

- Typically you use the long rows as power and ground
- Keep in mind rows are linked horizontally, and do not cross the middle divider
- There is also no connection from upper to lower rows

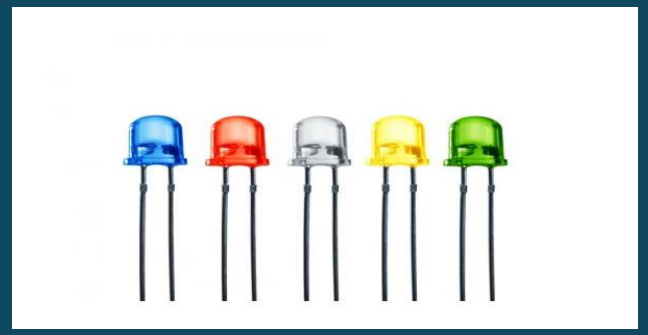



**NOTE: If you are using motors and high power consumption components, make sure you are careful with power ratings + consumption!!!**

# Some common components:

Motors

LEDs



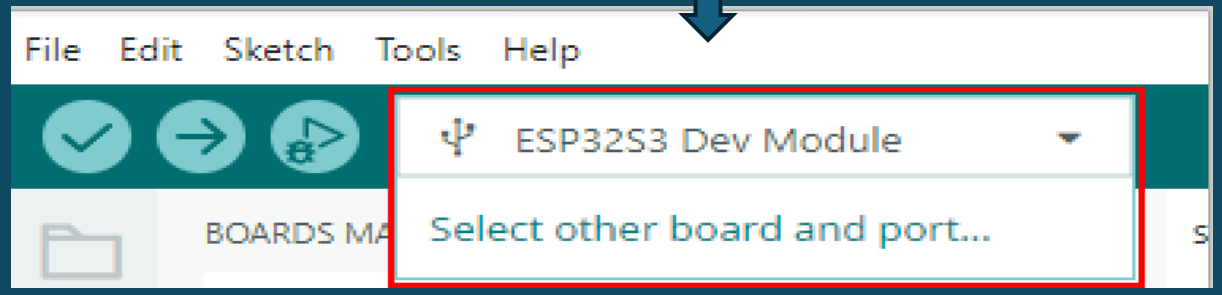
Buzzers

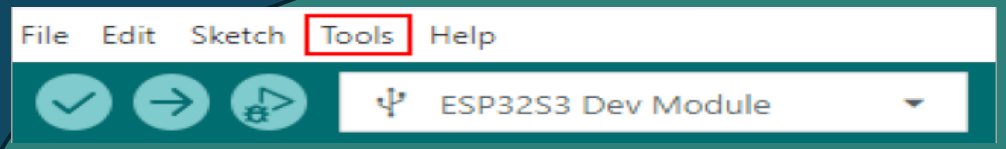
Sensors

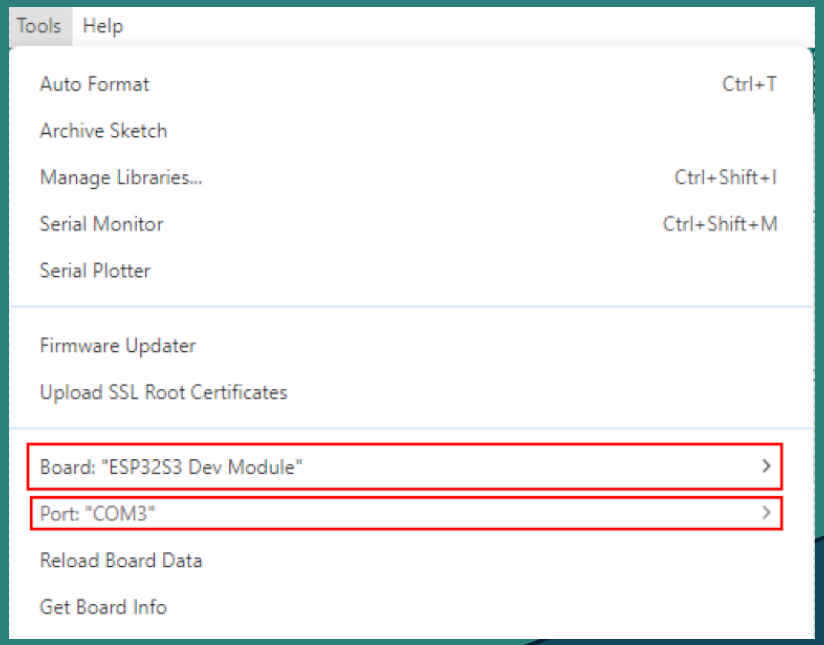
Potentiometers / Switches / Buttons

# Board + Port Setup

- To choose a port go to
  - Tools > Port > [select the port your arduino is connected to]
- To choose a board go to
  - Tools > Board > Arduino AVR Boards > Arduino Uno
- **(Or alternately select both from this dropdown)**



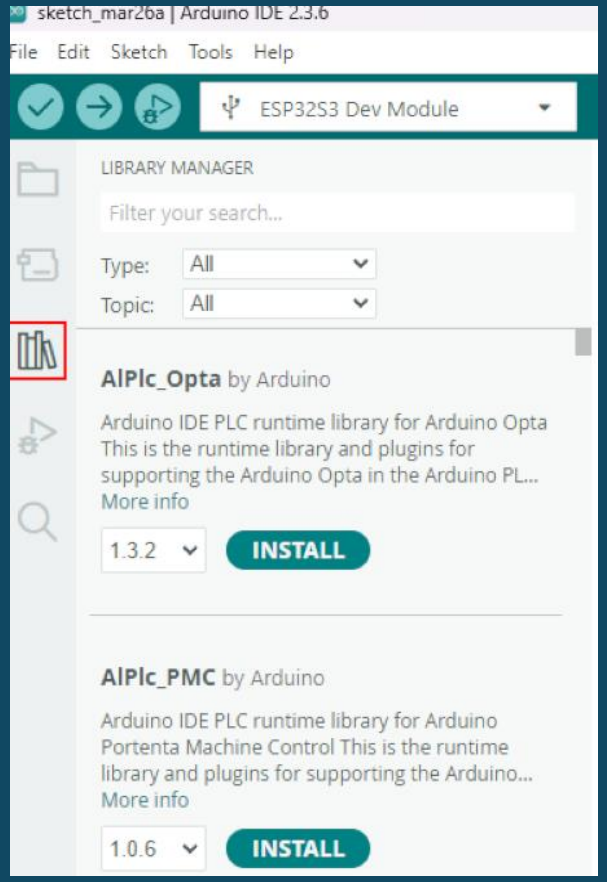



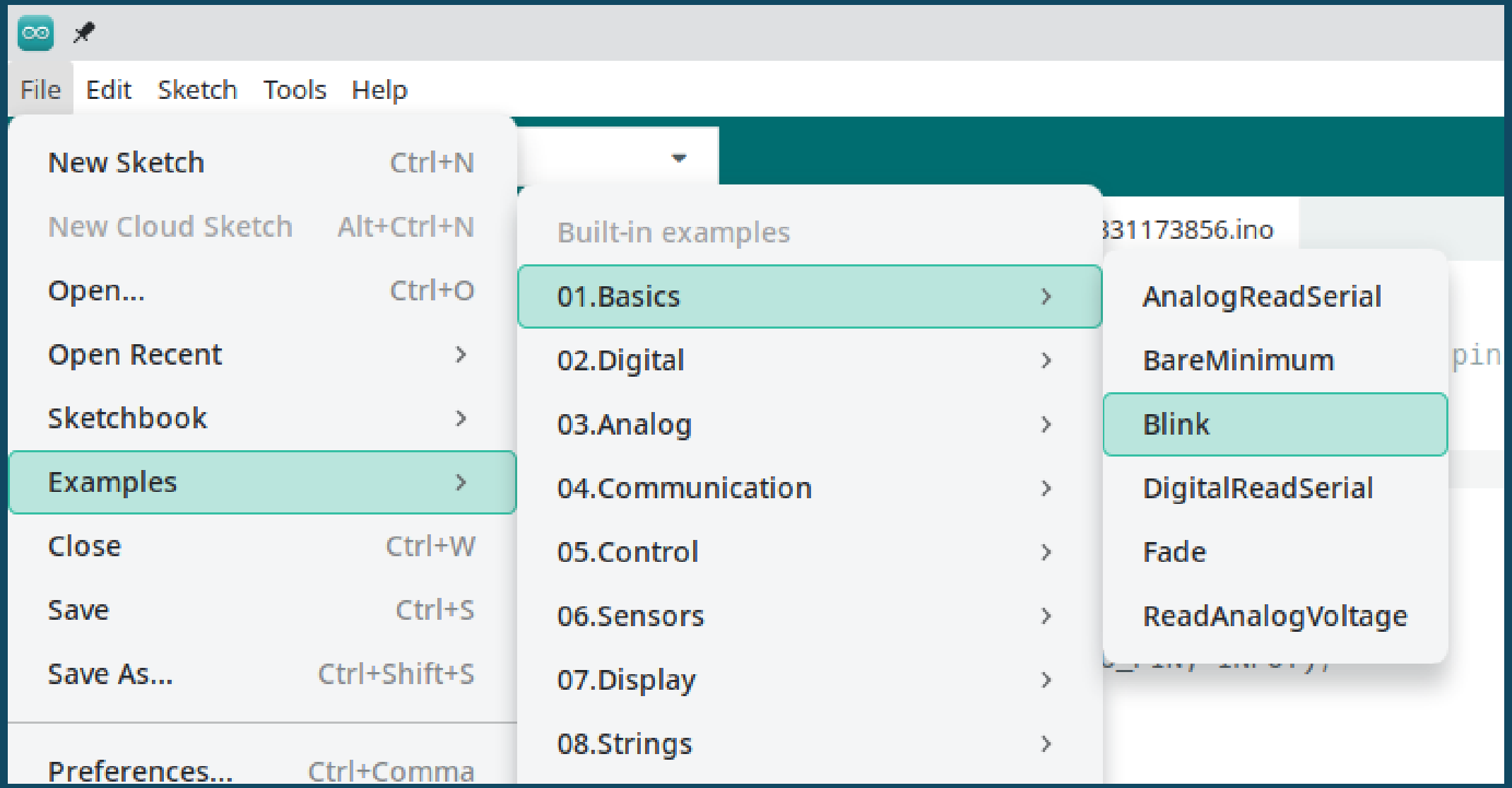
# FYI: Installing Libraries

- Libraries: Prebuilt code. They can provide specific functions for individual Arduino components.

- You can either install zip libraries or select in the Library Manager



# Basics of the Arduino IDE



# Basics of the Arduino IDE

- The IDE uses a language similar to **C++**
- Every program:
  - **Setup** function
  - **Loop** function
- Order of execution:
  - Setup()
  - Loop()
  - Loop()
  - Loop()
  - .....

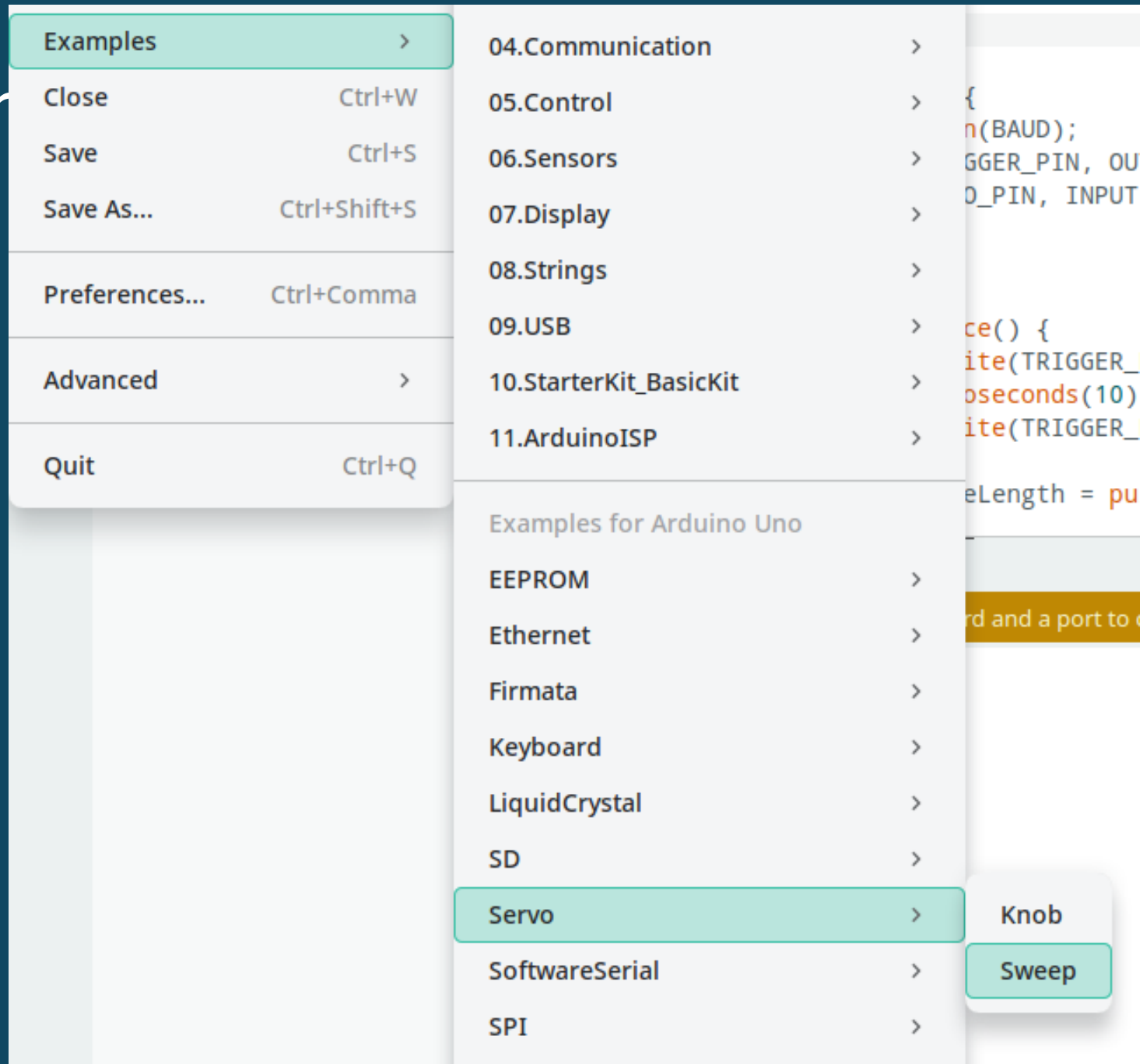
Blink.ino

```
void setup() {
  pinMode(LED_BUILTIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_BUILTIN, HIGH);
  delay(1000);
  digitalWrite(LED_BUILTIN, LOW);
  delay(1000);
}

```

# Basics of the Arduin



# Basics of the Arduino ID

- Plug in servo:
  - Ground (Brown)
  - 5V
  - Data (pin 3)
- Run the code
- Keep the servo plugged in

Sweep.ino readme.md

```
#include <Servo.h>

Servo myservo;
int pos = 0;

void setup() {
  myservo.attach(3);
}

void loop() {
  for (pos = 0; pos <= 180; pos += 1) {

    myservo.write(pos);
    delay(15);
  }
  for (pos = 180; pos >= 0; pos -= 1) {
    myservo.write(pos);
    delay(15);
  }
}
```

# Questions?

We will be moving on to the small project next.

# The project

- First, a short video about the components.
- https://www.youtube.com/watch?v=KGwtit2bFyo
- We will not be making the project in the video together.

# The project (pt. 1/2): Distance detection

- We will start from the "Blink" example!
  1. Connect the ultrasonic sensor
  2. Set up Serial printing ("Hello, World")
  3. Read and print the distance value

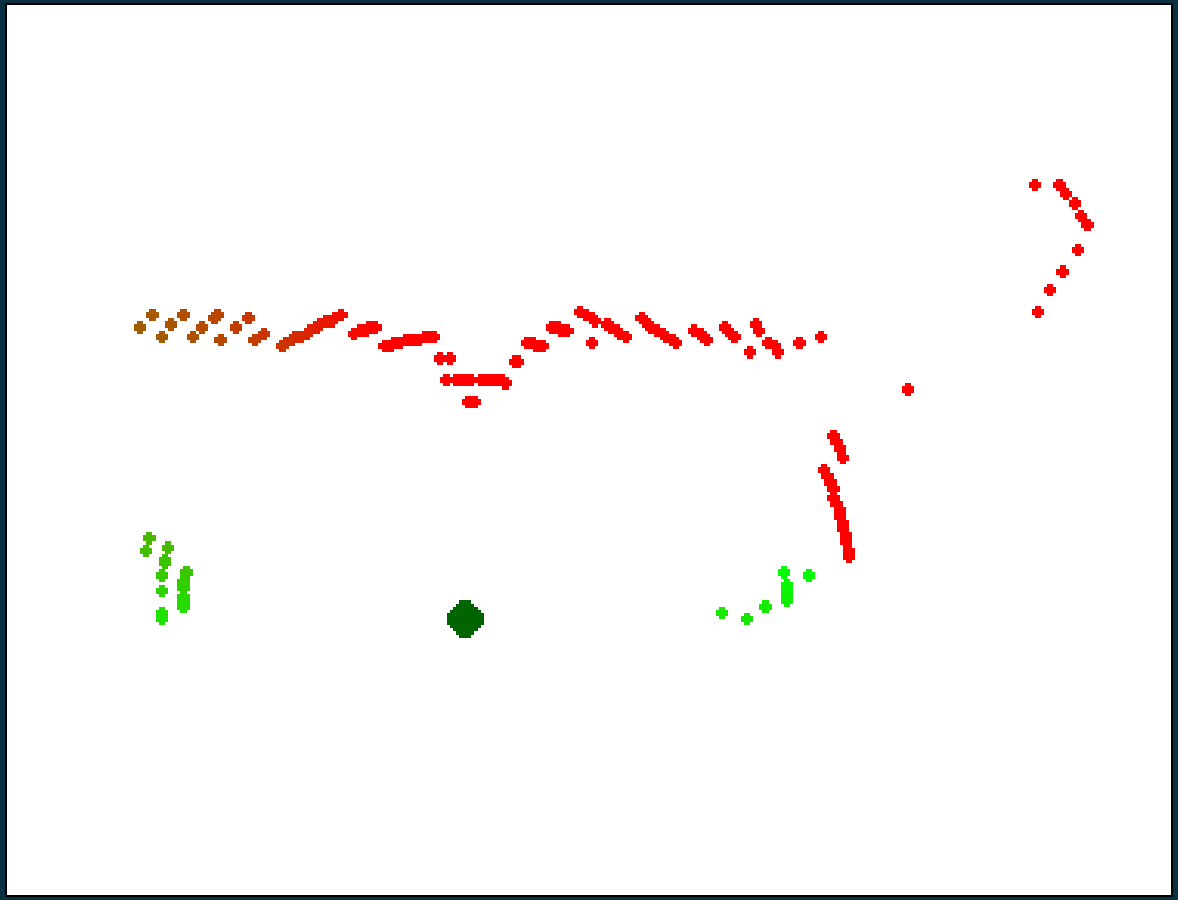
# The project (pt. 2/2): Distance Controlled Motor

1. Copy the code from 'Sweep' to set up the servo
2. Make sure your servo is still plugged in
3. Add the line of code:

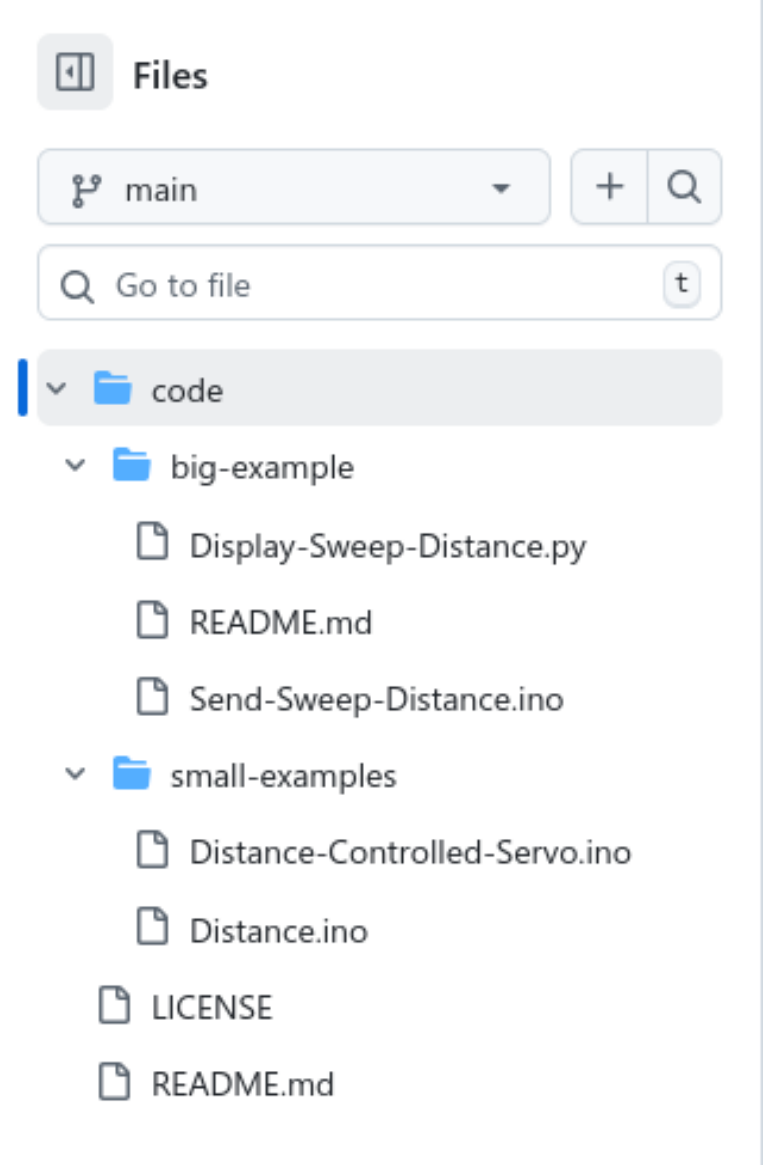
```
myServo.write(distance)
```

You're done!

# Big Example



# Code for all previous parts

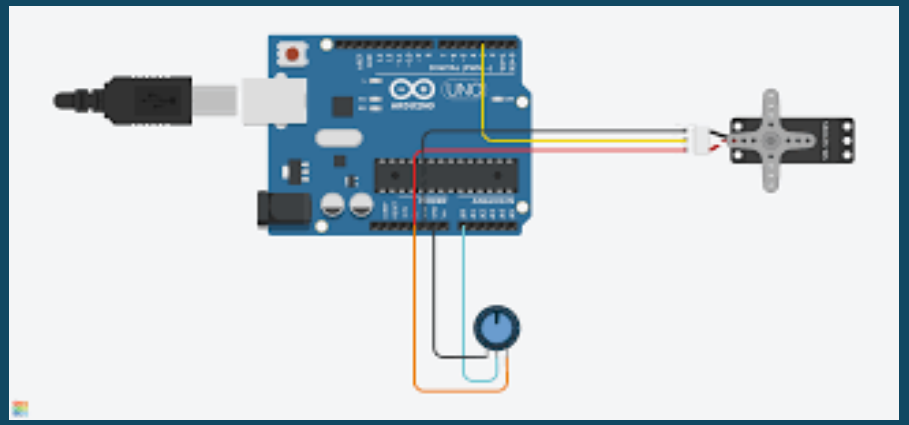



https://github.com/UMRoboticsTeam/arduino-workshop-April-2026/

# Short additions to this project
# If we have time… Try one!

- Easy:
  - Make the built-in LED turn on if the distance is close enough
  - If you've done that: Make an array of LEDs turn on one at a time the closer you get
- Medium:
  - Modify the 'Sweep' code to change the *speed* depending on the distance
- Hard:
  - Plug in the 'joystick' and use it to control the servo
    - The joystick resting position should keep the servo still
    - Left/right should control the servo
  - Note: The joystick has two potentiometers! Use analog read.

# Here are some resources to help you in the future



- **Tinkercad:** https://www.tinkercad.com/ : Free simulation

- These are some websites with premade projects:
  - https://projecthub.arduino.cc/
  - https://www.instructables.com/

- Our good friend Chat

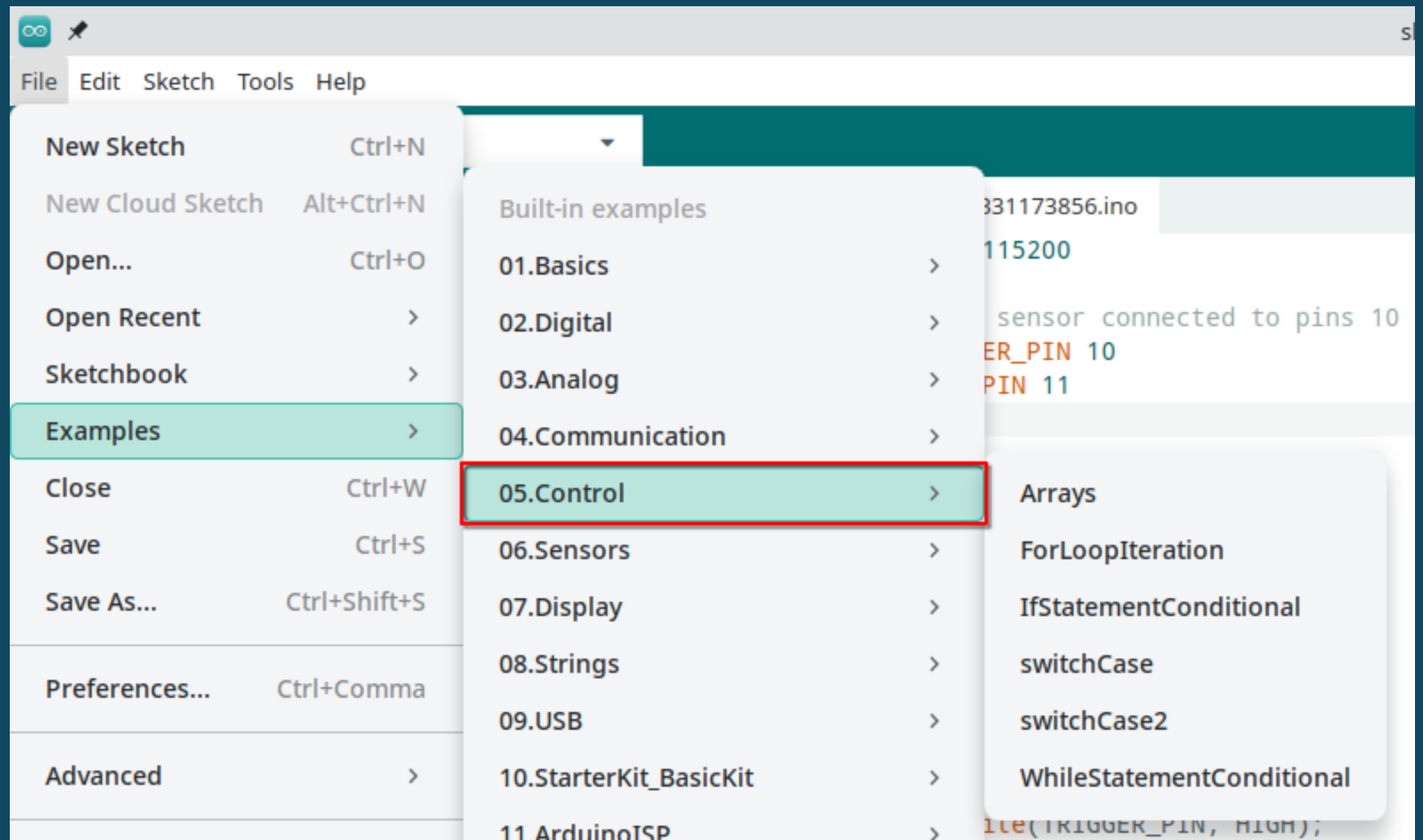
- **Starter kit [50$]:** https://a.co/d/09PPf3wS

# Tips from the '''expert'''

- VS Code Extension called "PlatformIO" lets you program Arduinos
  - This way you don't *need* the Arduino IDE. (look up a tutorial)
- ESP32 is (basically) Arduino with WiFi/bluetooth (wireless)
- If you buy pre-made sensors (like AdaFruit) they may have a library
- The Arduino examples are helpful to
  - See the correct syntax
  - Straight up copy
  - See how to use a specific library / tool (like a servo)

# Tips from the '''expert'''

- Helpful for the language

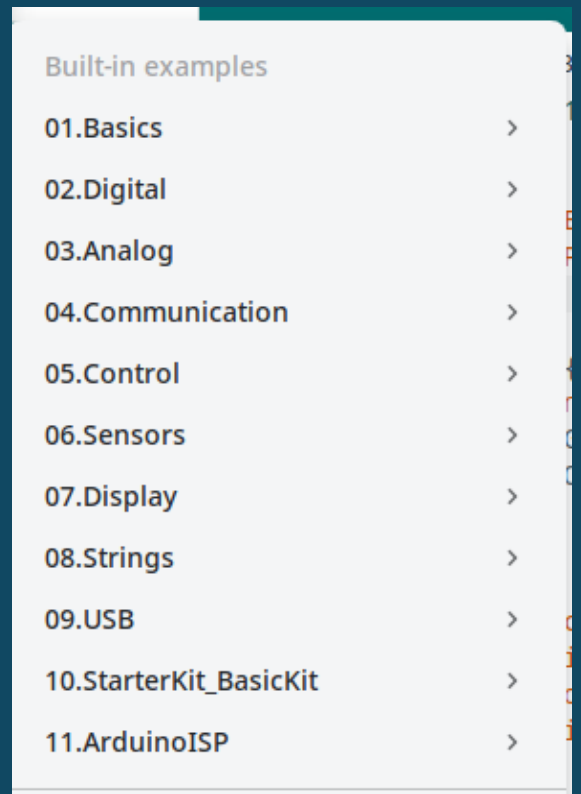


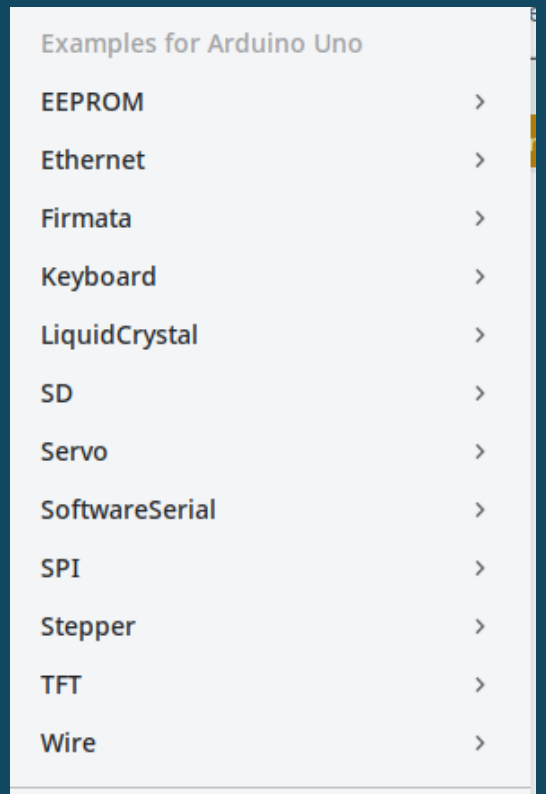
# Tips from the '''expert'''

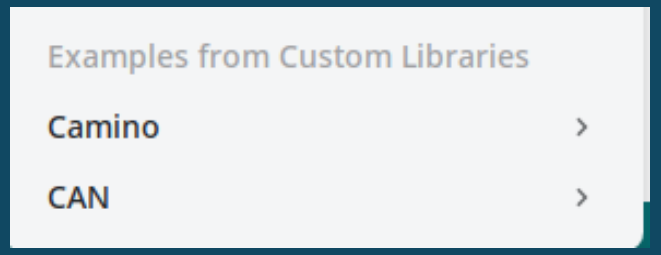
- Three categories of examples:
  - Built in (most basic)

For YOUR board (can be quite specific)

For YOUR libraries (you install; very specific)



# Thanks for joining us in the workshop!

You guys since you joined the workshop

If you have further questions, feel free to contact either of us on discord!!